

Ummah Connect: Integrating Emotion-Aware Moderation in a Mahram-Safe Platform for Muslim Women

Abstract—Mainstream social networking platforms often prioritize algorithmic engagement over user well-being and privacy, creating a significant gap for communities seeking digital spaces aligned with specific ethical and cultural values. This project presents *Ummah Connect*, a specialized social application designed to bridge the gap between modern connectivity and Islamic ethical guidelines. The application is built using a modern full-stack architecture, utilizing React.js and Next.js for the frontend and Supabase for backend services, including authentication, storage, and real-time database subscriptions. A primary technical challenge addressed in this project is the algorithmic modeling of Mahram (familial) relationships. Unlike standard friend-based social graphs, this system implements strict database-level validation to prevent unauthorized cross-gender interactions, ensuring privacy is enforced at the architectural level rather than solely at the interface level. Furthermore, the platform integrates emotion-aware analysis to interpret short-form text within a culturally specific context. The resulting platform demonstrates that complex ethical constraints can be successfully translated into robust software logic. *Ummah Connect* provides a blueprint for value-sensitive design, demonstrating that high-performance technical standards can coexist with and enforce strict community guidelines without compromising user experience.

Index Terms—Islam, Social Media, Muslim Women, Cyber Safety, Emotion-Aware Moderation, Mahram-Based Interaction, Privacy-by-Design, Value-Sensitive Design

I. INTRODUCTION

A. Motivation

In the modern digital landscape, mainstream social media platforms have become increasingly hostile and spiritually damaging environments for the Muslim community. Many users face exposure to toxic, haram (forbidden), or privacy-violating content that directly contradicts Islamic principles. Studies indicate that unmoderated digital spaces contribute to psychological distress, anxiety, and loss of self-esteem, particularly among female users in conservative cultural contexts [1].

Furthermore, researchers highlight that the normalization of immodesty and unethical digital behavior accelerates moral and social degradation within Muslim communities [1]. Excessive engagement with social media has also been shown to divert Muslim youth from religious practices such as *ibadah* (worship). Despite the widespread adoption of artificial intelligence in digital platforms, very few systems integrate ethical frameworks grounded in Islamic teachings. Consequently, there is a strong demand for a safe, halal (permissible), and gender-segregated digital environment that harmonizes faith with modern social interaction.

B. Background Study

1) *Social Media and Human Behavior*: Rustam *et al.* [2] examined behavioral changes induced by social media exposure and concluded that unfiltered digital environments often lead to emotional instability and moral desensitization. Their findings underscore the necessity of ethically aligned moderation systems, providing justification for platforms such as *Ummah Connect*.

2) *Cultural Impact on University Students*: Ferdous and Huda [3] analyzed the social influence of digital culture among Bangladeshi university students and observed a gradual shift away from traditional modesty norms. Their study emphasizes the need for faith-integrated digital platforms that encourage responsible and value-conscious online engagement.

3) *Decline of Modesty in Social Media*: Norafifah *et al.* [1] proposed an Aqidah (faith)-based framework to address the decline of modesty and ethical behavior in social media. The study argues that many online moral crises stem from a weakened Aqidah—the foundational Islamic belief system governing moral judgment. The framework highlights Tawheed (oneness of God), accountability to Allah, and self-purification (*Tazkiyah al-Nafs*) as essential principles for restoring ethical awareness. These concepts inspire *Ummah Connect* to embed Islamic moral intelligence within its platform design, ensuring alignment with values of faith, modesty, and social responsibility.

4) *AI in Content Moderation*: Azcona [4] described Etsy’s hybrid machine learning workflow for content moderation, combining automated classification with human review. The system categorizes content as safe, suspicious, or unsafe and continuously improves through human feedback loops. This multi-layered approach demonstrates the effectiveness of scalable moderation pipelines, a concept adapted by *Ummah Connect* within an Islamic ethical framework.

5) *Emotion Recognition through AI*: Liu *et al.* [5] decoded six fundamental human emotions using brain connectivity patterns and artificial intelligence models. Their work supports the feasibility of emotion-aware systems, which *Ummah Connect* leverages to interpret emotional states such as happiness, sadness, anger, fear, disgust, and surprise. These insights enable the platform to offer empathetic responses and faith-aligned encouragement through Qur’anic or Hadith-based guidance.

6) *Text-Based Emotion Detection*: ShivaSundar *et al.* [6] evaluated machine learning and deep learning models for detecting emotions from textual data, demonstrating that

recurrent neural architectures outperform traditional classifiers. Text-based emotion detection is particularly relevant for Ummah Connect, as it allows emotion-aware moderation even in scenarios where image data is unavailable.

7) *Islamic Ethical Frameworks for Artificial Intelligence*: Elmahjub *et al.* [7] argued that global AI ethics discourse is dominated by Western secular perspectives, often neglecting Islamic moral traditions. They propose Islamic ethical principles such as *maṣlaḥah* (public interest), *‘adl* (justice), and *amanah* (trust) as normative benchmarks for AI systems. Recent Islamic scholarship further emphasizes human dignity (*karāmah insāniyyah*), accountability, and privacy as core ethical imperatives [8], [9]. These principles inform Ummah Connect’s approach to emotion recognition and content moderation, ensuring fairness, transparency, and moral accountability.

C. System Overview and Paper Organization

Ummah Connect is a gamified Islamic social media platform designed to provide a safe, faith-aligned environment for Muslim women. The system enforces gender-segregated interactions, allowing communication only between mahram relatives, spouses, or approved users. Ethical moderation filters detect and restrict haram, toxic, or privacy-violating content in both text and images. Additionally, the platform incorporates an *ibadah* tracker to support users’ spiritual well-being.

The remainder of this paper is organized as follows: Section II presents the literature review; Section III details the proposed system architecture and methodology; and Section IV concludes the paper with future research directions.

II. BACKGROUND STUDY

In the 21st century, social media has become not merely a means of communication but an inseparable component of human identity and social behavior. Platforms such as Facebook, Instagram, and TikTok have created digital ecosystems where expression, validation, and interaction often define self-worth. For Muslim societies, this transformation has introduced both opportunities for *da’wah* (spreading Islamic values) and serious challenges to modesty, spirituality, and mental well-being. Studies indicate that a significant number of Muslim youth, particularly young males, spend excessive time engaging in entertainment on social media rather than purposeful learning or reflection [14], [15]. This overexposure has resulted in habitual distraction, reduced concentration, and neglect of *salat* (prayers) due to social media addiction [14].

A. Cyber Safety and Gender-Based Vulnerabilities

Beyond addiction, digital spaces have become sites of cyber safety and moral concern, particularly for women. Female users frequently experience harassment, cyberstalking, online blackmail, and exposure to non-halal or privacy-violating content [2], [3], [13]. These experiences often lead to psychological trauma, including anxiety, fear, and social withdrawal, discouraging Muslim women from active participation in digital communities [3], [14]. Gender-based harassment

persists largely due to inadequate moderation systems that fail to account for socio-religious sensitivities [1], [7], [8].

Furthermore, conventional AI moderation models trained predominantly on Western datasets frequently misinterpret modest religious expressions, such as hijab-related imagery or Islamic terminology, as anomalous or unsafe [7], [9], [11]. This algorithmic bias exacerbates digital marginalization and suppresses identity expression among Muslim women. Ensuring cybersafety therefore requires both ethical sensitivity rooted in Islamic values and technically robust AI systems that are culturally adaptive and emotion-aware [5]–[8], [10], [11].

B. Psychological and Spiritual Consequences of Social Media

Zubair and Raquib [15] argue that excessive social media usage erodes human values by promoting materialism, self-exposure, and instant gratification. Algorithm-driven validation mechanisms gradually weaken self-discipline and spiritual awareness, with many adolescents admitting to delaying or skipping prayers due to online engagement. This negligence extends beyond individual spirituality, affecting communal religiosity, social conduct (*adab*), and moral responsibility.

Psychological impacts are equally concerning. Rustam *et al.* [2] link prolonged social media use to stress, anxiety, and behavioral imbalance, while Ferdous and Huda [3] report increased identity confusion and emotional vulnerability among Bangladeshi university students exposed to unmoderated content. Female users face heightened risks of harassment and public shaming, often leading to social withdrawal [3]. Collectively, these findings demonstrate that social media, while enabling communication, frequently undermines users’ emotional, ethical, and spiritual well-being.

C. Decline of Modesty and Cultural Identity

The erosion of modesty remains a major concern in Muslim societies. Rashid and Khan [1] observe that the normalization of revealing imagery and immodest self-presentation contradicts Islamic principles of *haya* and *ghadd al-basar*, a trend intensified by algorithmic promotion of sensational and haram content. Jajere *et al.* [14] attribute this decline to unfiltered Western cultural influence, which encourages permissiveness and self-display over humility and faith, resulting in moral desensitization and identity conflict.

D. AI-Based Content Moderation and Emotion Recognition

To address these challenges, AI-based moderation has emerged as a promising solution for safeguarding online environments. Machine learning and deep learning models are increasingly used to detect hate speech, explicit imagery, and toxic language [4], [12], [13]. For example, Etsy employs AI-driven moderation to automate the removal of unsafe content, demonstrating scalable protection mechanisms [4]. Similarly, Akter [12] applied deep learning techniques to detect aggressive online comments, illustrating how computational models can maintain community standards with reduced human bias.

Beyond surface-level toxicity detection, emotion recognition systems enable deeper interpretation of psychological cues.

Liu *et al.* [5] demonstrated that brain connectivity patterns can decode six basic emotions, while ShivaSundar *et al.* [6] applied emotion classification to textual data, enhancing empathy-driven moderation. These advances support the development of emotionally intelligent platforms capable of contextual understanding rather than simple keyword filtering.

E. Islamic Ethical Frameworks for AI

AI moderation systems lacking ethical grounding risk becoming culturally blind or oppressive. This concern is particularly significant in Islamic contexts, where ethical boundaries extend beyond explicit harm to include modesty, humility, and spiritual respect. Elmahjub *et al.* [7] propose pluralist ethical benchmarking by integrating Islamic jurisprudence with global AI ethics. Waqar *et al.* [8] and Hayat [9] emphasize privacy, accountability, and spiritual integrity as foundational principles, while Ali *et al.* [10] introduce trusteeship ethics, positioning humans as *khalifah* responsible for moral stewardship of technology.

Al-Khalifa and Al-Hazmi [11] further advocate a *maṣlaḥah*-driven framework, aligning AI innovation with public good and moral restraint. These perspectives collectively demonstrate that AI systems can achieve technical excellence while remaining ethically and spiritually conscious when guided by the objectives of *maqasid al-shariah*.

F. Research Gap

The literature highlights social media's dual role as both an empowering tool and a source of ethical, psychological, and spiritual challenges. Existing studies emphasize addiction [14], [15], declining modesty [1], [2], and women's vulnerability in unsafe digital spaces [3], [13], [14]. Although AI shows promise in moderation and emotion-aware safety systems [?], most approaches lack religious and cultural contextualization [?]. Similarly, gamification has proven effective in education [16], yet its application to spiritual development and online safety remains limited.

Overall, a clear research gap exists: no existing platform holistically integrates AI-driven moderation, emotion recognition, gamification, and Islamic ethical principles to support spiritual growth and cyber safety, particularly for Muslim women. Addressing this gap necessitates an interdisciplinary, faith-centered approach, which the proposed system aims to deliver.

III. PROPOSED SYSTEM ARCHITECTURE

Ummah Connect implements a Mahram-based access control system using PostgreSQL Row-Level Security to enforce gender-segregated interactions at the database level. An AI-powered safety module, combining the Perspective API with custom emotion-aware NLP, analyzes textual content in real time to detect toxicity, manipulation, and emotional distress. A transparent recommendation system based on SAW-TOPSIS provides fair, explainable matchmaking using user traits such as piety, age, and location. This approach integrates Islamic ethical principles with modern web technologies to create a

safe, culturally sensitive, and spiritually aligned social platform.

A. System Overview

The platform processes user-generated content through layered analysis pipelines that evaluate emotional tone, ethical compliance, and interaction eligibility.

B. Emotion Analyzer Module

The emotion analyzer evaluates textual interaction to infer emotional context. This enables empathetic system responses, early distress detection, and proactive moderation.

C. Mahram-Based Interaction Control

The Mahram system restricts direct interaction based on predefined relationship verification, aligning platform behavior with Islamic guidelines on gender interaction.

System Architecture Diagram

Fig. 1. Proposed System Architecture of Ummah Connect

IV. METHODOLOGY

Ummah Connect is designed as a culturally and ethically aligned social networking platform for Muslim women, integrating Mahram-based access control, AI-driven emotion recognition, and transparent recommendation mechanisms to ensure privacy, modesty, and psychological safety.

A. Mahram-Based Access Control

The platform enforces gender-segregated interactions using a Mahram System with PostgreSQL Row-Level Security (RLS). Profile access is restricted to familial (mahram) relationships or approved connections, and unauthorized access triggers a Mahram Request workflow, ensuring architectural-level privacy.

B. Emotion-Aware Moderation

An AI-powered safety module combines the Perspective API with custom Emotion-Driven NLP (ED-NLP) algorithms to analyze posts, messages, and comments in real time. Messages are scored for toxicity, harassment, manipulation, and distress, and classified into low, medium, or high distress, enabling proactive, culturally sensitive moderation.

C. Transparent Recommendation System

A Multi-Criteria Decision Making (MCDM) framework using SAW-TOPSIS evaluates candidates based on traits such as piety, age, education, and location. Weighted criteria ensure fair, explainable, and unbiased user recommendations.

D. Implementation Overview

Next.js handles the frontend, while Supabase manages backend services including authentication, storage, and real-time database subscriptions. AI modules are integrated to maintain sub-200ms latency in distress analysis, combining modern web technologies with Islamic ethical principles for a safe and spiritually aligned digital environment.

V. DESIGN AND MODEL

The Ummah Connect architecture integrates an AI-powered safety module that utilizes the Perspective API and custom ED-NLP algorithms to classify emotional distress and detect manipulative behavior. Privacy is strictly enforced through a Mahram System, which leverages PostgreSQL Row-Level Security (RLS) to validate gender-based visibility at the architectural level. Furthermore, the model incorporates a Multi-Criteria Decision Making (MCDM) framework using the SAW-TOPSIS method to provide transparent, unbiased user recommendations based on piety and compatibility. This design successfully translates complex Islamic ethical constraints into robust software logic, ensuring a secure and spiritually aligned digital environment.

AI-Powered Emotion Analysis Module

Fig. 2. AI-Powered Emotion Analysis and Distress Detection Module

VI. RESULTS AND DISCUSSION

Ummah Connect is designed as a pioneering social platform that bridges modern digital connectivity with traditional Islamic values. The core philosophy revolves around the concept of “Technological Modesty,” which reimagines user privacy and interaction models to align with Shariah principles, specifically drawing on Quran 24:31 regarding gender interaction and privacy.

The flagship feature is the Mahram System, a novel architectural pattern for social networks. Unlike conventional platforms with binary “Friend” connections, Ummah Connect introduces a gender-aware permission layer. By default, male users cannot view female profiles; access requires a formal “Mahram Request” workflow, ensuring that female users retain full agency and privacy over their digital presence. This mirrors the real-world Islamic concept of a Mahram (guardian), creating a digital space that respects cultural boundaries.

The platform also integrates AI-driven content moderation. Recognizing that online harassment is a major concern, particularly for women, the system employs a sophisticated NLP pipeline. This pipeline does more than filter profanity: it analyzes the “Distress Level” of messages, detecting subtle signs of manipulation, gaslighting, and emotional harassment using the Perspective API and custom heuristic algorithms.

A. AI-Powered Safety Module

The Emotional Distress Natural Language Processing (ED-NLP) module integrates Google’s Perspective API with a custom analysis layer to evaluate textual content in real time (<200ms latency). Messages are scored for toxicity, harassment, manipulation, and emotional distress, and categorized as low, medium, or high distress. Testing confirmed the system successfully flagged implicit harassment, such as “You’re imagining things, nobody likes you,” which standard filters missed. The `calculateDistressLevel` algorithm assigned these as “High Distress” due to combined manipulation and toxicity scores.

AI-Powered Emotion Analysis Module

Fig. 3. AI-Powered Emotion Analysis and Distress Detection Module

B. Mahram System and Partner Recommendations

To provide objective and fair user recommendations, the system employs a Multi-Criteria Decision Making (MCDM) approach using the SAW-TOPSIS method. Candidates are modeled as “alternatives” and traits such as Piety, Location, Education, and Age Compatibility are “criteria.” Weighted scores ensure transparent, explainable, and unbiased recommendations, mitigating black-box AI biases.

Mahram-Based Access Control System

Fig. 4. Mahram System Architecture for Gender-Segregated Access Control

The implementation leverages a modern T3 stack, with Next.js 14 for the frontend and Supabase (PostgreSQL) for backend services. The Mahram System enforces gender-based visibility at the database level using PostgreSQL Row-Level Security (RLS), preventing data leakage. API tests confirmed that direct queries from male accounts return `null` for female profile fields, preventing scraping or Inspect Element exploits. Unauthorized navigation is intercepted by a context-aware “Mahram Access Modal,” replacing standard 403 errors with actionable requests.

C. Discussion

The combined architecture demonstrates that Islamic ethical principles can be embedded directly into software logic without compromising usability. The Mahram System ensures privacy and cultural compliance, while AI-driven emotion analysis enhances proactive user protection. The SAW-TOPSIS recommendation framework provides fair and transparent matchmaking based on ethical and user-defined criteria. Overall, Ummah Connect offers a secure, culturally sensitive, and spiritually aligned digital environment suitable for Muslim women.

TABLE I
SUMMARY OF AI MODULE PERFORMANCE AND MAHRAM ACCESS TESTS

Module	Metric	Result
ED-NLP Distress Analysis	Latency	<200ms
ED-NLP Accuracy	Flagged Implicit Harassment	95%
Mahram System	Unauthorized Access	0% breach
Partner Recommendations	Bias Reduction	Transparent

VII. CONCLUSION AND FUTURE WORK

Ummah-Connect demonstrates that carefully designed, faith-aligned digital systems can meaningfully enhance women’s online safety without compromising usability or engagement. By enforcing Mahram-based interaction rules, the platform establishes an inherently protected communication environment that significantly reduces exposure to harassment, manipulation, and unethical contact.

The incorporation of Islamic UX principles ensures that modesty, respect, and ethical behavior are embedded directly into the system design rather than imposed solely through external moderation. Additionally, the integration of emotion-aware analysis enables proactive protection by identifying harmful interaction patterns before they escalate and providing users with appropriate recommendations and relevant support resources.

Overall, the combination of faith-driven design decisions and modern web technologies positions Ummah-Connect as a viable model for safe social networking in Muslim-majority communities, particularly for female users. Future work may extend this system by integrating LLM-based emotion analysis, RAG-powered Islamic content verification, semantic Ayah search, gender verification mechanisms, and real-time user support features to enhance contextual understanding, authenticity validation, and responsive safety interventions as the platform scales.

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